

SoCalGuessr Image Classification

DSC140B Final Project

March 17, 2026

1 Human Baseline

I performed a 50-image test using the SoCalGuessr platform. I was tasked with classifying images from six Southern California cities and achieved a score of **9/50 (18% accuracy)**. The confusion matrix in Figure 1 illustrates a heavy bias toward misclassifying various cities as Anaheim or Riverside, likely due to regional architectural similarities.

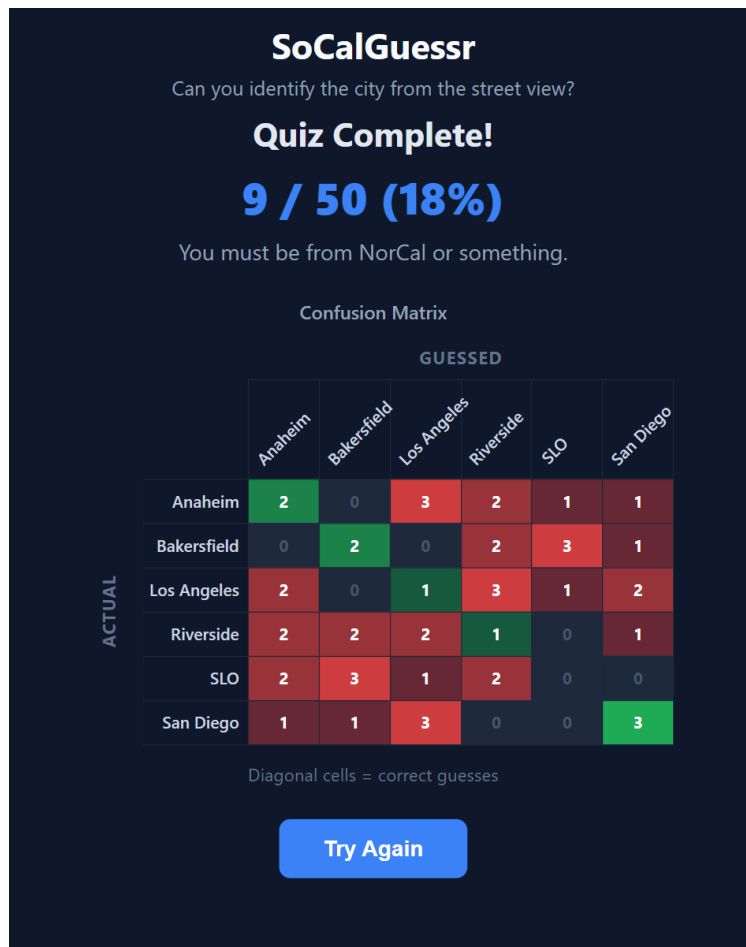


Figure 1: Human Baseline Confusion Matrix: 18% overall accuracy.

2 Final Model Architecture

The final model utilizes a MobileNetV3-Small backbone, replacing the last layer with a 6-class classifier.

2.1 Layer Configuration and Units

- **Backbone:** The architecture consists of **11 Inverted Residual (Bneck) blocks**.
- **Feature Map:** The backbone terminates in a final feature expansion layer of **576 units**.
- **Classifier (Head):** The model contains **0 hidden layers** in the classifier. The 576 features are mapped directly to the 6 output classes via a single `nn.Linear` layer.

2.2 Backbone Architecture Specifics

The model utilizes the **MobileNetV3-Small** feature extractor, optimized for high-efficiency mobile vision. Each Bneck block follows a Narrow-Wide-Narrow structure:

1. **Expansion:** 1×1 pointwise convolution.
2. **Depthwise Convolution:** 3×3 or 5×5 spatial filtering.
3. **Squeeze-and-Excitation (SE):** Attention mechanism to focus on informative features.
4. **Projection:** 1×1 convolution back to a lower-dimensional bottleneck.

2.3 Detailed Backbone Layer Specifications

Input	Operator	Expansion Size	Channels	SE	NL	Stride
$224^2 \times 3$	Conv2d, 3×3	-	16	-	HS	2
$112^2 \times 16$	Bneck, 3×3	16	16		RE	2
$56^2 \times 16$	Bneck, 3×3	72	24	-	RE	2
$28^2 \times 24$	Bneck, 3×3	88	24	-	RE	1
$28^2 \times 24$	Bneck, 5×5	96	40		HS	2
$14^2 \times 40$	Bneck, 5×5	240	40		HS	1
$14^2 \times 40$	Bneck, 5×5	240	40		HS	1
$14^2 \times 40$	Bneck, 5×5	120	48		HS	1
$14^2 \times 48$	Bneck, 5×5	144	48		HS	1
$14^2 \times 48$	Bneck, 5×5	288	96		HS	2
$7^2 \times 96$	Bneck, 5×5	576	96		HS	1
$7^2 \times 96$	Bneck, 5×5	576	96		HS	1
$7^2 \times 96$	Conv2d, 1×1	-	576	-	HS	1

Table 1: MobileNetV3-Small Backbone Configuration.

2.4 Final Parameter Audit

Parameter Type	Count
Non-Trainable (Frozen)	927,008
Trainable	596,998
Total Parameters	1,524,006

Table 2: Parameter distribution for the final model.

3 Training Procedure

The model was trained using the **Adam** optimizer and **Cross Entropy Loss**. The training process took approximately **1 hour and 50 minutes**.

3.1 Training Curve and Empirical Risk

As shown in Figure 2, the training loss decreased significantly over 20 epochs. While training accuracy reached near 99%, validation accuracy peaked at approximately 84%.

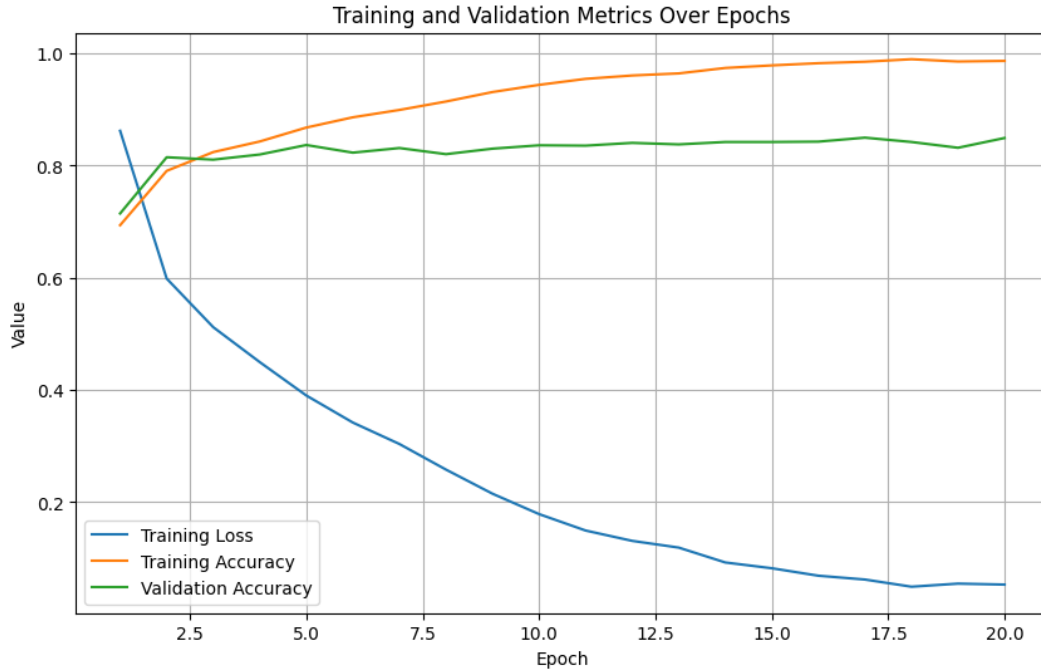


Figure 2: Training and Validation Metrics Over 20 Epochs.